

Practical - 9

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BOOKFLOW LIBRARY SYSTEM

Transactional Microservice using Spring Boot, Spring Data JPA and PostgreSQL

To develop a transactional BookFlow Library microservice using Spring Boot, Spring Data JPA and PostgreSQL for managing book inventory and borrowing orders, with FastAPI Gateway integration.

Database – bookflow creation in Postgresql

```
postgres=# CREATE DATABASE bookflow;
CREATE DATABASE
postgres=# \c bookflow
You are now connected to database "bookflow" as user "postgres".
bookflow=# \dt
Did not find any tables.
```

Tables Inserted

```
bookflow=# \dt
          List of tables
 Schema |   Name   | Type  | Owner
-----+-----+-----+-----
 public | inventory | table | postgres
 public | orders   | table | postgres
(2 rows)
```

Initial Inventory Data in PostgreSQL

```
bookflow=# INSERT INTO inventory
bookflow-# (book_id, total_stock, available_stock)
bookflow-# VALUES
bookflow-# (101, 5, 5),
bookflow-# (102, 3, 3),
bookflow-# (103, 2, 0);
INSERT 0 3
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock
-----+-----+-----
    101  |              5  |          5
    102  |              3  |          3
    103  |              0  |          2
(3 rows)
```

GET Inventory for Book 101

```
2026-10-01T12:02:30.250+05:30 INFO 23396 --- [demo] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
Hibernate:
    select
        i1_0.book_id,
        i1_0.available_stock,
        i1_0.total_stock
    from
        inventory i1_0
    where
        i1_0.book_id=?
```

localhost:8080/inventory/101

Pretty print

```
{
  "bookId": 101,
  "totalStock": 5,
  "availableStock": 5
}
```

Successful Book Borrowing — POST /orders

POST http://localhost:8080/orders

Docs Params Authorization Headers 8 Body Scripts Settings

none form-data x-www-form-urlencoded raw binary G

```
1 {
2   "userId": 501,
3   "bookId": 101
4 }
```

Body Cookies Headers 5 Test Results

{ } JSON Preview Visualize

```
1 {
2   "id": 1,
3   "userId": 501,
4   "bookId": 101,
5   "status": "BORROWED"
6 }
```

GET /inventory/101 — Inventory After Book

Borrowing

```
Hibernate:
select
  i1_0.book_id,
  i1_0.available_stock,
  i1_0.total_stock
from
  inventory i1_0
where
  i1_0.book_id=?
```

localhost:8080/inventory/101

Pretty print

```
{
  "bookId": 101,
  "totalStock": 5,
  "availableStock": 4
}
```

PostgreSQL Database — Inventory and Order Records After

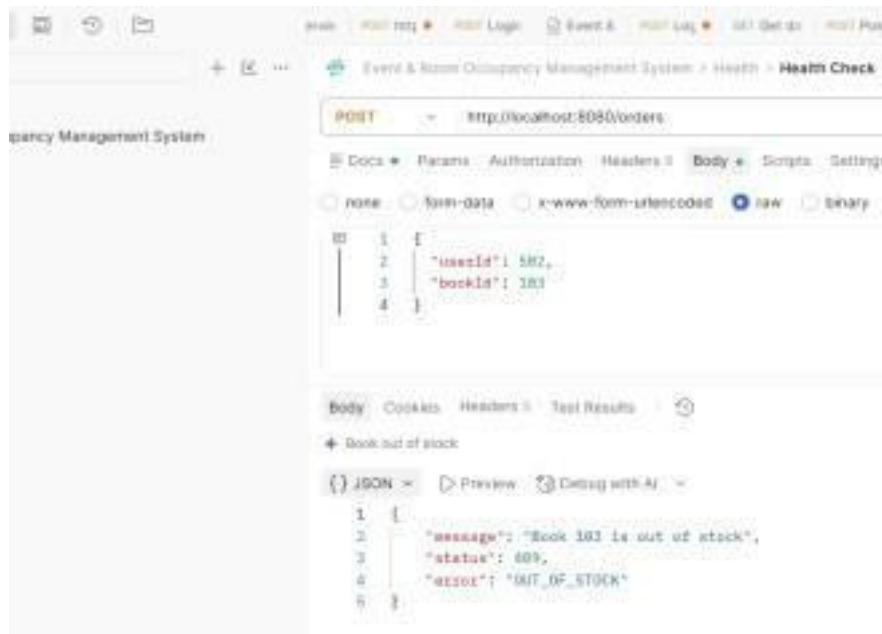
```
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock
-----+-----+-----
    101  |              5 |          5
    102  |              3 |          3
    103  |              0 |          2
(3 rows)

bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock
-----+-----+-----
    102  |              3 |          3
    103  |              0 |          2
    101  |              4 |          5
(3 rows)

bookflow=# SELECT * FROM orders;
 id | book_id | status  | user_id
-----+-----+-----+-----
  1 |    101 | BORROWED |     501
(1 row)
```

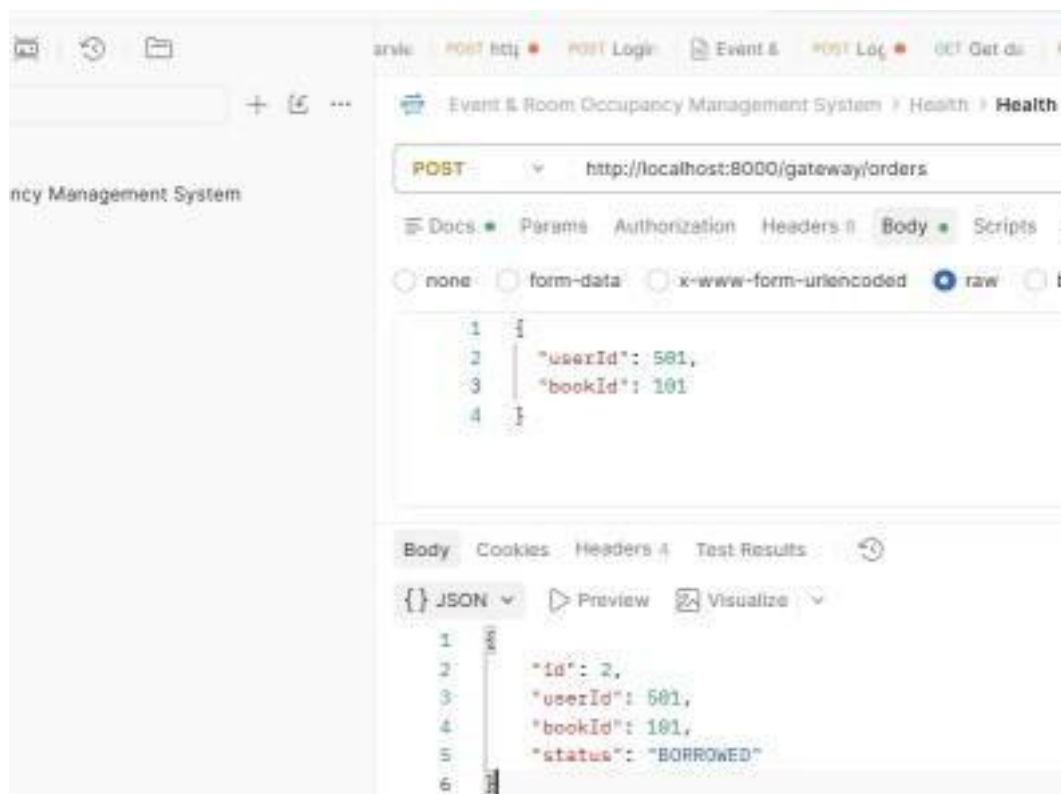
Borrowing

POST /orders — Out-of-Stock Book Returns 409 Conflict



FastAPI Gateway — /gateway/orders Successfully Forwards Request to Spring Boot

```
INFO: 127.0.0.1:64794 - "POST /gateway/orders HTTP/1.1" 200 OK
```



PostgreSQL — Final Inventory and Order Records

```
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock
-----+-----+-----
      102 |                3 |           3
      103 |                0 |           2
      101 |                3 |           5
(3 rows)
```

```
bookflow=# SELECT * FROM orders;
 id | book_id | status  | user_id
----+-----+-----+-----
  1 |    101 | BORROWED |    501
  2 |    101 | BORROWED |    501
(2 rows)
```

Folder Structure:

bookflow-transaction/

```
|
|
|— pom.xml
|
|
|— src/
|   |
|   |— main/
|   |   |
|   |   |— java/
|   |   |   |
|   |   |   |— com/
|   |   |   |   |
|   |   |   |   |— bookflow/
|   |   |   |   |   |
|   |   |   |   |   |— transaction/
|   |   |   |   |   |   |
|   |   |   |   |   |   |— BookflowTransactionApplication.java
|   |   |   |   |   |   |
|   |   |   |   |   |   |
|   |   |   |   |   |   |— controller/
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |— OrderController.java
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |— InventoryController.java
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |— entity/
```

```

| | └── Inventory.java
| | └── Order.java
| |
| | └── repository/
| |   ├── InventoryRepository.java
| |   └── OrderRepository.java
| |
| | └── service/
| |   ├── InventoryService.java
| |   └── OrderService.java
| |
| | └── dto/
| |   └── OrderRequest.java
| |
| | └── exception/
| |   ├── BookOutOfStockException.java
| |   └── GlobalExceptionHandler.java
| |
| | └── resources/
| |   └── application.yml

```

Codes:

· pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>

```

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>3.5.5</version>
<relativePath/>
</parent>

<groupId>com.bookflow</groupId>
<artifactId>bookflow-transaction</artifactId>
<version>0.0.1-SNAPSHOT</version>

<name>bookflow-transaction</name>
<description>BookFlow Transactional Microservice</description>

<properties>
  <java.version>17</java.version>
</properties>

<dependencies>

  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
  </dependency>

  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
  </dependency>

  <dependency>
    <groupId>org.postgresql</groupId>
    <artifactId>postgresql</artifactId>
    <scope>runtime</scope>
  </dependency>

  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-validation</artifactId>
  </dependency>

  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-test</artifactId>
    <scope>test</scope>
  </dependency>
```

```

</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
      <configuration>
        <mainClass>com.bookflow.transaction.BookflowTransactionApplication</main
inClass>
      </configuration>
    </plugin>
  </plugins>
</build>

</project>

```

· application.yml

```

spring:
  datasource:
    url: jdbc:postgresql://localhost:5432/bookflow
    username: postgres
    password: UR_PASSWORD

```

```

jpa:
  hibernate:
    ddl-auto: update
    show-sql: true
  properties:
    hibernate:
      format_sql: true

```

```

server:
  port: 8080

```

· BookflowTransactionApplication.java

```

package com.bookflow.transaction;

```

```

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

```

```
@SpringBootApplication
public class BookflowTransactionApplication {

    public static void main(String[] args) {
        SpringApplication.run(BookflowTransactionApplication.class, args); }
}
· Inventory.java
```

```
package com.bookflow.transaction.entity;
```

```
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import jakarta.validation.constraints.Min;
```

```
@Entity
@Table(name = "inventory")
public class Inventory {

    @Id
    private Long bookId;

    @Min(value = 0, message = "Total stock cannot be negative")
    private Integer totalStock;

    @Min(value = 0, message = "Available stock cannot be negative") private
    Integer availableStock;

    public Inventory() {
    }

    public Inventory(Long bookId, Integer totalStock, Integer availableStock) {
        this.bookId = bookId;
        this.totalStock = totalStock;
        this.availableStock = availableStock;
    }

    public Long getBookId() {
        return bookId;
    }

    public void setBookId(Long bookId) {
        this.bookId = bookId;
    }
}
```

```

    public Integer getTotalStock() {
        return totalStock;
    }

    public void setTotalStock(Integer totalStock) {
        this.totalStock = totalStock;
    }
    public Integer getAvailableStock() {
        return availableStock;
    }

    public void setAvailableStock(Integer availableStock) {
        this.availableStock = availableStock;
    }
}

```

· Order.java

```

package com.bookflow.transaction.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name = "orders")
public class Order {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private Long userId;
    private Long bookId;
    private String status;

    public Order() {
    }

    public Order(Long userId, Long bookId, String status) { this.userId
        = userId;
        this.bookId = bookId;
    }
}

```

```

        this.status = status;
    }

    public Long getId() {
        return id;
    }

    public void setId(Long id) {
        this.id = id;
    }

    public Long getUserId() {
        return userId;
    }

    public void setUserId(Long userId) {
        this.userId = userId;
    }

    public Long getBookId() {
        return bookId;
    }

    public void setBookId(Long bookId) {
        this.bookId = bookId;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }
}

```

· InventoryRepository.java

```

package com.bookflow.transaction.repository;

import com.bookflow.transaction.entity.Inventory;
import org.springframework.data.jpa.repository.JpaRepository;

public interface InventoryRepository extends JpaRepository<Inventory, Long> { }

```

· OrderRepository.java

```
package com.bookflow.transaction.repository;

import com.bookflow.transaction.entity.Order;
import org.springframework.data.jpa.repository.JpaRepository;

public interface OrderRepository extends JpaRepository<Order, Long> { }
```

· OrderRequest.java

```
package com.bookflow.transaction.dto;

import jakarta.validation.constraints.NotNull;

public class OrderRequest {

    @NotNull
    private Long userId;

    @NotNull
    private Long bookId;

    public Long getUserId() {
        return userId;
    }

    public void setUserId(Long userId) {
        this.userId = userId;
    }

    public Long getBookId() {
        return bookId;
    }

    public void setBookId(Long bookId) {
        this.bookId = bookId;
    }
}
```

· InventoryService.java

```
package com.bookflow.transaction.service;
```

```

import com.bookflow.transaction.entity.Inventory;
import com.bookflow.transaction.repository.InventoryRepository; import
org.springframework.stereotype.Service;

@Service
public class InventoryService {

    private final InventoryRepository inventoryRepository;

    public InventoryService(InventoryRepository inventoryRepository) {
        this.inventoryRepository = inventoryRepository;
    }
    public Inventory getInventory(Long bookId) {
        return inventoryRepository.findById(bookId)
            .orElseThrow(() -> new RuntimeException("Book not found")); }
}

```

· OrderService.java

```

package com.bookflow.transaction.service;

import com.bookflow.transaction.dto.OrderRequest;
import com.bookflow.transaction.entity.Inventory;
import com.bookflow.transaction.entity.Order;
import com.bookflow.transaction.exception.BookOutOfStockException; import
com.bookflow.transaction.repository.InventoryRepository; import
com.bookflow.transaction.repository.OrderRepository; import
org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
public class OrderService {

    private final InventoryRepository inventoryRepository;
    private final OrderRepository orderRepository;

    public OrderService(
        InventoryRepository inventoryRepository,
        OrderRepository orderRepository) {
        this.inventoryRepository = inventoryRepository;
        this.orderRepository = orderRepository;
    }
}

```

```

@Transactional
public Order borrowBook(OrderRequest request) {

    Inventory inventory = inventoryRepository.findById(request.getBookId())
        .orElseThrow(() -> new RuntimeException("Book not found"));

    if (inventory.getAvailableStock() <= 0) {
        throw new BookOutOfStockException(
            "Book " + request.getBookId() + " is out of stock");
    }

    inventory.setAvailableStock(
        inventory.getAvailableStock() - 1);

    inventoryRepository.save(inventory);
    Order order = new Order(
        request.getUserId(),
        request.getBookId(),
        "BORROWED");

    return orderRepository.save(order);
}
}

```

· InventoryController.java

```

package com.bookflow.transaction.controller;

import com.bookflow.transaction.entity.Inventory;
import com.bookflow.transaction.service.InventoryService; import
org.springframework.web.bind.annotation.*;

```

```

@RestController
@RequestMapping("/inventory")
public class InventoryController {

    private final InventoryService inventoryService;

    public InventoryController(InventoryService inventoryService) {
        this.inventoryService = inventoryService;
    }

    @GetMapping("/{bookId}")
    public Inventory getInventory(@PathVariable Long bookId) { return

```

```

        inventoryService.getInventory(bookId);
    }
}

```

· OrderController.java

```

package com.bookflow.transaction.controller;

import com.bookflow.transaction.dto.OrderRequest;
import com.bookflow.transaction.entity.Order;
import com.bookflow.transaction.service.OrderService; import
jakarta.validation.Valid;
import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/orders")
public class OrderController {
    private final OrderService orderService;

    public OrderController(OrderService orderService) {
        this.orderService = orderService;
    }

    @PostMapping
    @ResponseStatus(HttpStatus.CREATED)
    public Order borrowBook(@Valid @RequestBody OrderRequest request) { return
        orderService.borrowBook(request);
    }
}

```

· BookOutOfStockException.java

```

package com.bookflow.transaction.exception;

public class BookOutOfStockException extends RuntimeException {

    public BookOutOfStockException(String message) {
        super(message);
    }
}

```

· GlobalExceptionHandler.java

```

package com.bookflow.transaction.exception;

```

```

import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.*;

import java.util.Map;

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(BookOutOfStockException.class)
    @ResponseStatus(HttpStatus.CONFLICT)
    public Map<String, Object> handleOutOfStock(
        BookOutOfStockException ex) {

        return Map.of(
            "status", 409,
            "error", "OUT_OF_STOCK",
            "message", ex.getMessage()
        );
    }
}

```

· gateway.py — FastAPI gateway

```

from fastapi import FastAPI
import httpx

app = FastAPI()

SPRING_SERVICE = "http://localhost:8080"

@app.post("/gateway/orders")
async def create_order(order: dict):
    async with httpx.AsyncClient() as client:
        response = await client.post(
            f"{SPRING_SERVICE}/orders",
            json=order
        )
    return response.json()

```